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ORIGINAL PAPER

Backtesting VaR and expectiles with realized scores

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Abstract
Several statistical functionals such as quantiles and expectiles arise naturally as the minimizers of the expected value of a scoring function, a property that is called elicitability (see Gneiting in J Am Stat Assoc 106:746–762, 2011 and the references therein). The existence of such scoring functions gives a natural way to compare the accuracy of different forecasting models, and to test comparative hypotheses by means of the Diebold–Mariano test as suggested in a recent work. In this paper we suggest a procedure to test the accuracy of a quantile or expectile forecasting model in an absolute sense, as in the original Basel I backtesting procedure of value-at-risk. To this aim, we study the asymptotic and finite-sample distributions of empirical scores for normal and uniform i.i.d. samples. We compare on simulated data the empirical power of our procedure with alternative procedures based on empirical identification functions (i.e. in the case of VaR the number of violations) and we find an higher power in detecting at least misspecification in the mean. We conclude with a real data example where both backtesting procedures are applied to AR(1)–Garch(1,1) models fitted to SP500 logreturns for VaR and expectiles' forecasts.

Keywords Backtesting · Forecasting · Value at risk · Expectiles

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1 Introduction

The notion of backtesting a risk measure has been the subject of a lot of research either in the academia and in the financial industry. Roughly, backtesting means developing a statistical test of a forecasting model, based on the sequence of realized portfolio losses and of risk measure forecasts. The aim of backtesting is to reject the forecasting model if the chosen test statistics, that should be related with the risk measure under scrutiny, is too far from its theoretical value.

Perhaps the first and most well-known example of backtesting procedure is the Basel I methodology for backtesting value-at-risk (VaR henceforth) by means of the sequence of violations. Recall that VaR at level α is simply the α-quantile of the forecasting distribution and a violation is said to occur if the realized loss is bigger than VaR. Under the null hypothesis that the forecasting model is correct the sequence of violations is Bernoulli i.i.d. with probability α; classical backtesting procedures are based on counting the number of violations and/or measuring their tendency to cluster (see Kupiec 1995; Christoffersen 1998; Christoffersen and Pelletier 2004, or Campbell 2005 for a review).

One common critique to traditional backtesting procedures of VaR is that taking into account only the number of violations and not their magnitude lacks statistical power and may be misleading. For example, a risk manager could easily get a Bernoulli i.i.d. sequence of violations and thus pass a backtesting based on the number of violations by issuing extremely conservative VaR forecasts with probability $1-\alpha$ and extremely permissive VaR forecast with probability α in a random fashion. This argument has been further explored in Holzmann and Eulert (2014), that have shown that traditional backtesting of VaR is insensitive with respect to increasing information.

The idea of backtesting VaR using also the magnitude of violations already emerged in Lopez (1999a, b), but never became popular among the practitioners. More recently, a similar approach has been pursued in Wong (2010), that used the mean of the magnitude of the violations as a backtesting statistic for VaR.

After the seminal paper of Artzner et al. (1999), academic research recognized some shortcomings of VaR and suggested to move to expected shortfall (ES) as a proper risk measure for computing capital requirements; the problem of backtesting ES has been considered for example by Macneil and Frey (2000), Kerkhof and Melenberg (2004) and Acerbi and Szekely (2014).

A paper that had a big influence and initiated an active stream of new research on backtesting was Gneiting (2011), where the important concepts of consistent scoring function and elicitable functional were introduced in the financial community. A scoring function is consistent for a given statistical functional if the functional can be defined as the minimizer of the expected value of the score; a functional is elicitable if it admits a consistent scoring function. Notice that scores are actually meant as losses, the smaller the better. Consistent scoring functions give a natural way to compare different forecasts of the same risk measure.

The simplest elicitable risk measures are VaR, the expectiles, and the couple (VaR, ES). Indeed, VaR is the minimizer of the expected value of a suitable piecewise linear score; expectiles are by definition the minimizers of a suitable piecewise-quadratic score; and it has been recently established by Acerbi and Szekely (2014) and Fissler and Ziegel (2016) that the couple (VaR, ES) jointly minimizes the expectations of a suitable family of scoring functions.

In the financial literature scoring functions have been used mainly for comparative backtesting, that is the analysis of the comparative merit of two different forecasting models. With comparative backtesting, introduced by Fissler et al. (2016) and Nolde and Ziegel (2016), it is possible to reject the null hypothesis that one forecasting model is better than another, with a procedure that is based on the Diebold–Mariano test on the difference of realized scores. Of course, comparative backtesting does not give any information about the validity of the considered forecasting models: one can compare extremely poor models without noticing.

In this paper we suggest to use scoring functions for traditional backtesting, in order to validate a model in an absolute sense, in contrast with the aforementioned comparative backtesting. The idea is pretty straightforward: since the true value of the forecasted elicitable functional minimizes the expected value of a scoring function, if the forecasting model is not too bad then the realized score should be not too big. In order to build a formal statistical test, we derive the asymptotic distribution of the realized score for normal and uniform i.i.d. samples. We focus here on the cases of VaR and expectiles, but the same ideas can also be applied for the couple (VaR, ES).

In order to apply backtesting with realized scores to more complex econometric models, it is necessary to derive the distribution of the realized scores under the null hypothesis that the forecasting model is correct. This can always be done by simulation, as it was suggested for expectiles in Bellini and Di Bernardino (2017). In this paper we complement the simulation approach with an alternative approach, based on the probability integral transform, as it has been suggested by Berkowitz (2001) for VaR and by Kerkhof and Melenberg (2004) for ES (see also Corbetta and Peri 2016).

The paper is structured as follows. In Sect. 2 we introduce the basic notations. In Sect. 3 we derive asymptotic distributions of realized scores and identification functions and study their small sample distribution by means of simulations. In Sect. 4 we provide a simple example on simulated data. In Sect. 5 we provide an example of backtesting standard AR(1)–Garch(1,1) models, while Sect. 6 concludes the paper with some remarks and suggestions for further developments.

2 Notations and preliminaries

We denote with Y the random future loss associated with a financial position. The α-Value at Risk of Y is simply the α-quantile

$$v_\alpha(Y) = \inf\{t : F_Y(t) \geq \alpha\},$$

where $F_Y(t) = P(Y \leq t)$ and typically $\alpha = 0.05$ or $\alpha = 0.01$. As it is well known from quantile regression (see e.g. Koenker, 2005),

$$v_\alpha(Y) = \arg\min_{x\in\mathbb{R}} E\left[S^{(v)}(x, Y)\right],$$

for the piecewise linear scoring function

$$S^{(v)}(x, y) = \alpha(y - x)_+ + (1 - \alpha)(y - x)_-. \tag{1}$$

Newey and Powell (1987) introduced the expectiles e_α as the minimizers of the expected value of an asymmetric quadratic scoring function: for $Y \in L^2$,

$$e_\alpha(Y) = \arg\min_{x \in \mathbb{R}} E\left[S^{(e)}(x, Y)\right],$$

with

$$S^{(e)}(x, y) = \alpha(y - x)_+^2 + (1 - \alpha)(y - x)_-^2. \tag{2}$$

Expectiles are thus an asymmetric generalization of the mean, that arise when $\alpha = 1/2$. Expectiles are becoming increasingly popular in the financial literature since they are the only elicitable coherent risk measures; for this characterization and further properties of expectiles we refer to Weber (2006), Bellini (2012), Bellini et al. (2014), Bellini and Bignozzi (2014), Bellini and Di Bernardino (2017) and Delbaen et al. (2016). It has to be stressed that the scoring functions $S^{(v)}$ and $S^{(e)}$ are by no means the only ones whose expected values are minimized respectively by the quantiles and the expectiles. Under mild technical conditions, Thomson (1979) and Saerens (2000) proved that the more general scoring function consistent with quantiles has the form

$$S(x, y) = \left(1_{\{x \le y\}} - \alpha\right)(g(x) - g(y)) \tag{3}$$

for some nondecreasing $g \colon \mathbb{R} \to \mathbb{R}$, while Gneiting (2011) showed that in the case of expectiles the more general consistent scoring function is

$$S(x, y) = \left|1_{\{x \le y\}} - \alpha\right| \left(\phi(y) - \phi(x) - \phi'(x)(y - x)\right), \tag{4}$$

where $\phi \colon \mathbb{R} \to \mathbb{R}$ is a convex function with right derivative ϕ'. Newey and Powell (1987) showed that for $Y \in L^1$ expectiles are the unique solutions of

$$E\left[I^{(e)}(x, Y)\right] = 0,$$

where the so called identification function $I^{(e)}$ is given by

$$I^{(e)}(x, y) = \alpha(y - x)_+ - (1 - \alpha)(y - x)_-. \tag{5}$$

By analogy with (5), we denote the identification function of quantiles with

$$I^{(v)}(x, y) = \alpha 1_{\{y > x\}} - (1 - \alpha) 1_{\{y < x\}}. \tag{6}$$

Let now $Y_k,\ k = 1, \dots, n$, be an i.i.d. sample. The empirical versions of expected scores and identification functions will be called respectively realized scores and realized identification functions and are defined as follows:

$$\widehat{S}_n^{(v)}(x) = \frac{1}{n}\sum_{k=1}^{n} S^{(v)}(x, Y_k), \qquad \widehat{S}_n^{(e)}(x) = \frac{1}{n}\sum_{k=1}^{n} S^{(e)}(x, Y_k),$$
$$\widehat{I}_n^{(v)}(x) = \frac{1}{n}\sum_{k=1}^{n} I^{(v)}(x, Y_k), \qquad \widehat{I}_n^{(e)}(x) = \frac{1}{n}\sum_{k=1}^{n} I^{(e)}(x, Y_k). \tag{7}$$

We notice that similar definitions may be given for other scoring functions of the families (3) and (4), although in this paper we stick to $S^{(v)}$ and $S^{(e)}$. Another possibility could be the simultaneous use of different scoring functions, following the idea of Murphy diagrams introduced in Ehm et al. (2016).

3 Asymptotic distributions of realized scores and identification functions

In this section we study the asymptotic distribution of the realized scores and identification functions (7) for normal and uniform i.i.d. samples. All the computations are reported in the Appendix. Let $Y \sim N(0, 1)$, and let ϕ, Φ and z_α be respectively its density, distribution function and α-quantile. We have

$$E\left[S^{(v)}(v_\alpha, Y)\right] = \phi(v_\alpha)$$
$$Var\left[S^{(v)}(v_\alpha, Y)\right] = \alpha(1-\alpha)\left(1 + v_\alpha^2\right) + (1-2\alpha)v_\alpha\phi(v_\alpha) - \phi^2(v_\alpha),$$

where $S^{(v)}$ has been defined in (1). If $Y \sim U[0, 1]$, we get

$$E\left[S^{(v)}(v_\alpha, Y)\right] = \frac{\alpha(1-\alpha)}{2}$$
$$Var\left[S^{(v)}(v_\alpha, Y)\right] = \frac{\alpha^2(1-\alpha)^2}{12}.$$

Some values of $E\left[S^{(v)}(v_\alpha, Y)\right]$ and $Var\left[S^{(v)}(v_\alpha, Y)\right]$ are reported in Table 1.

Moving to the expectile scoring function (2), in the normal case we get

$$E\left[S^{(e)}(e_\alpha, Y)\right] = (1-2\alpha)e_\alpha\phi(e_\alpha) + \left(1 + e_\alpha^2\right)\left[\alpha\overline{\Phi}(e_\alpha) + (1-\alpha)\Phi(e_\alpha)\right]$$
$$Var\left[S^{(e)}(e_\alpha, Y)\right] = \left[3 + 6e_\alpha^2 + e_\alpha^4\right]\left\{\alpha^2 - 2\alpha\Phi(e_\alpha) + \Phi(e_\alpha)\right\}$$
$$+ (1-2\alpha)\left(5e_\alpha\phi(e_\alpha) + e_\alpha^3\phi(e_\alpha)\right) - E^2\left[S^{(e)}(e_\alpha, Y)\right].$$

Table 1 Mean and variance of the realized score $S^{(v)}(v_\alpha, Y)$ in the normal (upper) and in the uniform (lower) case for different values of α

α	0.5	0.05	0.01	0.001
v_α	0	-1.6449	-2.3263	-3.0902
$E[S^{(v)}]$	3.9894×10^{-1}	1.0314×10^{-1}	2.6652×10^{-2}	3.3671×10^{-3}
$Var[S^{(v)}]$	9.0845×10^{-2}	1.2698×10^{-2}	2.0053×10^{-3}	1.4337×10^{-4}
v_α	0.5	0.05	0.01	0.001
$E[S^{(v)}]$	1.2500×10^{-1}	2.3750×10^{-2}	4.9500×10^{-3}	4.9950×10^{-4}
$Var[S^{(v)}]$	5.2083×10^{-3}	1.8802×10^{-4}	8.1675×10^{-6}	8.3167×10^{-8}

Recall that in the uniform case the expectile is given by

$$e_\alpha = \begin{cases} \frac{\alpha-\sqrt{\alpha-\alpha^2}}{2\alpha-1} & \text{if } \alpha \neq 1/2 \\ 1/2 & \text{if } \alpha = 1/2. \end{cases} \tag{8}$$

For $\alpha \neq 1/2$ we get

$$E\left[S^{(e)}(e_\alpha, Y)\right] = \frac{\alpha(1-\alpha)\left(1-2\sqrt{\alpha(1-\alpha)}\right)}{3\,(2\alpha-1)^2}$$
$$Var\left[S^{(e)}(e_\alpha, Y)\right] = \frac{4\alpha^2\,(1-\alpha)^2\left(1-2\sqrt{\alpha(1-\alpha)}\right)^2}{45\,(2\alpha-1)^4}.$$

If $\alpha = 1/2$ the expectile coincides with the mean, so we get $E\left[S^{(e)}(e_{1/2}, Y)\right] = 1/24$ and $Var\left[S^{(e)}(e_{1/2}, Y)\right] = 1/720$. Notice that for each $\alpha \in (0, 1)$

$$SD\left[S^{(e)}(e_\alpha, Y)\right] = \frac{2}{\sqrt{5}} E\left[S^{(e)}(e_\alpha, Y)\right],$$

that shows that for a uniform i.i.d. sample the coefficient of variation of the realized score does not depend on α. Some values of $E\left[S^{(e)}(e_\alpha, Y)\right]$ and $Var[S^{(e)}(e_\alpha, Y)]$ are reported in Table 2.

We now perform similar computations for the identification functions $I^{(v)}$ and $I^{(e)}$. From the definitions (5), (6), it holds that

$$E\left[I^{(v)}(Y, v_\alpha)\right] = 0$$
$$E\left[I^{(e)}(Y, v_\alpha)\right] = 0$$

Table 2 Mean and variance of the realized score $S^{(e)}$ in the normal (upper) and in the uniform (lower) case for different values of α

α	0.5	0.05	0.01	0.001
e_α	0	-1.1402	-1.7173	-2.4265
$E[S^{(e)}]$	0.5	1.6440×10^{-1}	5.2091×10^{-2}	8.4147×10^{-3}
$Var[S^{(e)}]$	0.5	8.6607×10^{-2}	1.6176×10^{-2}	1.4115×10^{-3}
e_α	0.5	1.8661×10^{-1}	9.1325×10^{-2}	3.0668×10^{-2}
$E[S^{(e)}]$	4.1667×10^{-2}	1.1027×10^{-2}	2.7523×10^{-3}	3.1320×10^{-4}
$Var[S^{(e)}]$	1.3889×10^{-3}	9.7273×10^{-5}	6.0601×10^{-6}	7.8476×10^{-8}

so $E[\widehat{I}_n^{(v)}] = 0$ and $E[\widehat{I}_n^{(e)}] = 0$. Moreover, notice that in the case of VaR

$$\begin{aligned}\widehat{I}_n^{(v)}(v_\alpha) &= \frac{1}{n}\sum_{k=1}^{n} I^{(v)}(v_\alpha, Y_k) = \frac{1}{n}\sum_{k=1}^{n} \alpha 1_{\{Y_k > v_\alpha\}} - (1-\alpha) 1_{\{Y_k < v_\alpha\}} \\ &= \frac{[\alpha(n - N_V) - (1-\alpha)N_V]}{n} = \alpha - \frac{N_V}{n},\end{aligned}$$

where

$$N_V = \sum_{k=1}^{n} 1_{\{Y_k < v_\alpha\}}$$

is the number of violations. So for VaR the realized identification function is a linear transformation of the number of violations. Hence the exact finite sample distribution of the realized identification function is binomial; in other words, in this case backtesting with the realized identification function coincides with the traditional backtesting with the number of violations and there is no need to use an asymptotic distribution since the exact finite distribution is known.

In the case of expectiles, if Y has a standard normal distribution, then

$$Var[I^{(e)}(Y, e_\alpha)] = \left(1 + e_\alpha^2\right)\left[\alpha^2 + \Phi(e_\alpha)(1-2\alpha)\right] + e_\alpha \phi(e_\alpha)\left[1 - 2\alpha\right],$$

while if Y has a uniform distribution

$$Var[I^{(e)}(Y, e_\alpha)] = \frac{\alpha^2 (1-e_\alpha)^3}{3} + \frac{(1-\alpha)^2 e_\alpha^3}{3}.$$

Some numerical values of $Var[I^{(e)}(Y, e_\alpha)]$ are reported in Table 3.

Under the assumption that Y_n are i.i.d., realized scores and realized identification functions are time averages of i.i.d. variables, so from the law of large numbers and the central limit theorem it follows that

Table 3 Variance of the realized score $I^{(e)}$ in the normal (upper) and in the uniform (lower) case for different values of α

α	0.5	0.05	0.01	0.001
e_α	0	-1.1402	-1.7173	-2.4265
$Var\left[I^{(e)}\right]$	2.5000×10^{-1}	5.5147×10^{-2}	1.2995×10^{-2}	1.5337×10^{-3}
e_α	0.5	1.8661×10^{-1}	9.1325×10^{-2}	3.0668×10^{-2}
$Var\left[I^{(e)}\right]$	–	2.4032×10^{-3}	2.7385×10^{-4}	9.8993×10^{-6}

$$\widehat{S}_n^{(v)}(x) \to E[S^{(v)}(x, Y)] \text{ a.s.},$$
$$\sqrt{n}\left(\widehat{S}_n^{v}(x) - E[S^{v}(x, Y)]\right) \xrightarrow{d} N(0, Var[S^{v}(x, Y)]),$$

and similarly for $S_n^{(e)}$, $I_n^{(v)}$, $I_n^{(e)}$. The whole point in constructing a backtesting procedure by means of realized scores or realized identification functions is to be able to compute explicitly asymptotic variances, as we did in Tables 1, 2 and 3. Notice also that asymptotic normality may hold under much more general hypotheses, related to the ergodicity of the process Y_n, although we do not explore this issue further in the present paper.

3.1 Simulations

In order to investigate the departure from normality of the finite sample distributions of $\widehat{S}_n^{(v)}$, $\widehat{S}_n^{(e)}$ and $\widehat{I}_n^{(e)}$, we perform a simulation experiment. The case of $\widehat{I}_n^{(v)}$ is left out because here the exact finite sample distribution is binomial, as we discussed in the previous subsection. We simulate $N = 100{,}000$ trajectories of lengths $n = 100$ and $n = 1000$, for $\alpha = 0.05, 0.01, 0.001$, and for each trajectory we compute $\widehat{S}_n^{(v)}$, $\widehat{S}_n^{(e)}$ and $\widehat{I}_n^{(e)}$. Normal expectiles have been computed by means of the R function `enorm`, while uniform expectiles by formula (8). The empirical distributions of the 100,000 realizations of $\widehat{S}_n^{(v)}$, $\widehat{S}_n^{(e)}$ and $\widehat{I}_n^{(e)}$ are reported respectively in Figs. 1, 2, 3, 4, 5 and 6. Odd-numbered figures refer to the case of a normal sample, while even-numbered figures refer to the case of a uniform sample. First of all, as it is expected, in all cases the quality of the approximation improves when moving from $n = 100$ to $n = 1000$. Moreover, the quality of the approximation is much worse for smaller values of α; the intuitive reason is that for a fixed Y, the distribution of $S^{(v)}(x, Y)$, $S^{(e)}(x, Y)$ and $I^{(e)}(x, Y)$ becomes highly asymmetric when α is very small, so the convergence to the normal limit is likely to be slower. Secondly, we see that the convergence is much better in the uniform case (Figs. 2, 4, 6) than in the normal case (Figs. 1, 3, 5). Indeed, from a graphical point of view we see that in the uniform case the normal approximation is acceptable in all cases, with the only exception of $\widehat{I}_n^{(e)}$, $n = 100$, $\alpha = 0.001$ (Fig. 6). Finally, the quality of the approximation seems slightly better in the case of $\widehat{S}_n^{(e)}$ than in the case of $\widehat{S}_n^{(v)}$ (Fig. 1 vs. Fig. 3), although this might be due to the specific distributions considered.

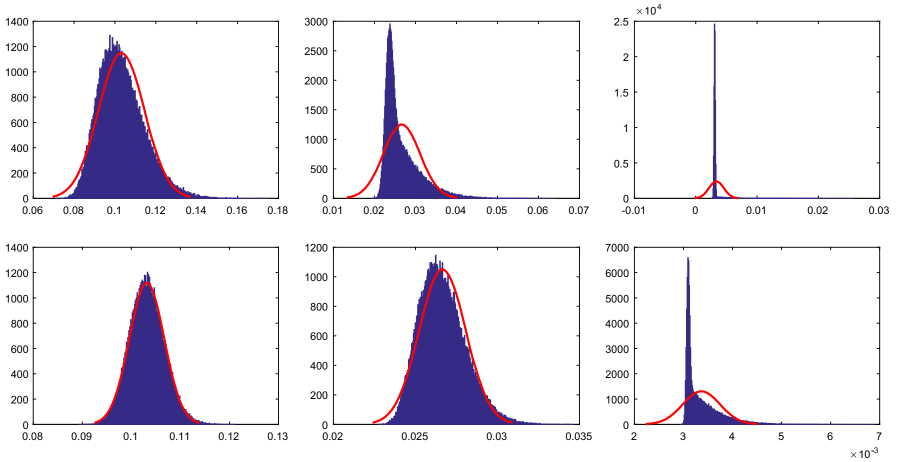

Fig. 1 Distributions of 100,000 realizations of the realized score $\widehat{S}_n^{(v)}$ for a normal i.i.d. sample with $n = 100$ (upper part) and $n = 1000$ (lower part), for $\alpha = 0.05, 0.01, 0.001$

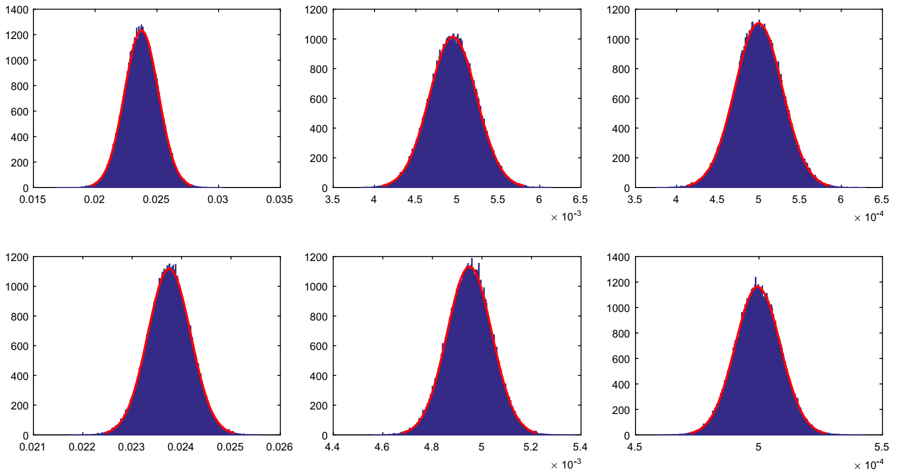

Fig. 2 Distributions of 100,000 realizations of the realized score $\widehat{S}_n^{(v)}$ for a uniform i.i.d. sample with $n = 100$ (upper part) and $n = 1000$ (lower part), for $\alpha = 0.05, 0.01, 0.001$

4 Backtesting with realized scores and identification functions

The aim of this section is to provide a first example of backtesting forecasted quantiles and forecasted expectiles by means of realized scores and realized identification functions on simulated data. Since scoring functions are minimized in the true value of quantiles and expectiles, the idea of the proposed tests is to reject the forecasting model if the realized score is too high.

Similarly, since the expected value of identification functions are equal to 0 for the true value of quantiles and expectiles, we will reject the forecasting model if the realized values of the identification functions are sufficiently far away from 0. As a

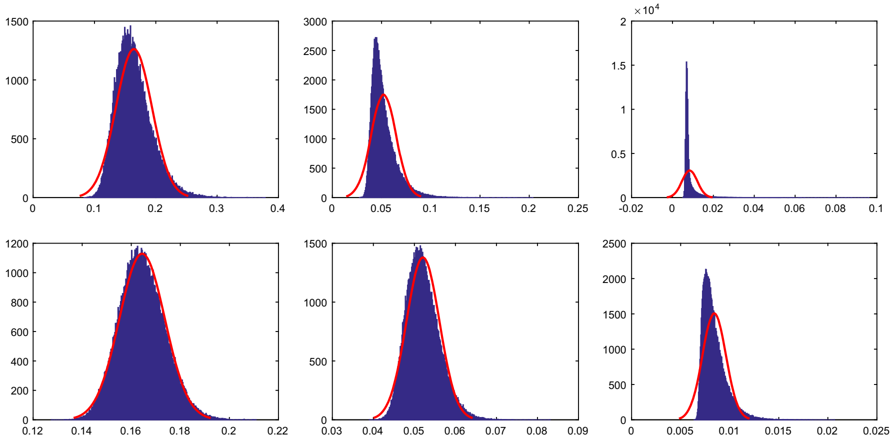

Fig. 3 Distributions of 100,000 realizations of the realized score $\widehat{S}_n^{(e)}$ for a normal i.i.d. sample with $n = 100$ (upper part) and $n = 1000$ (lower part), for $\alpha = 0.05, 0.01, 0.001$

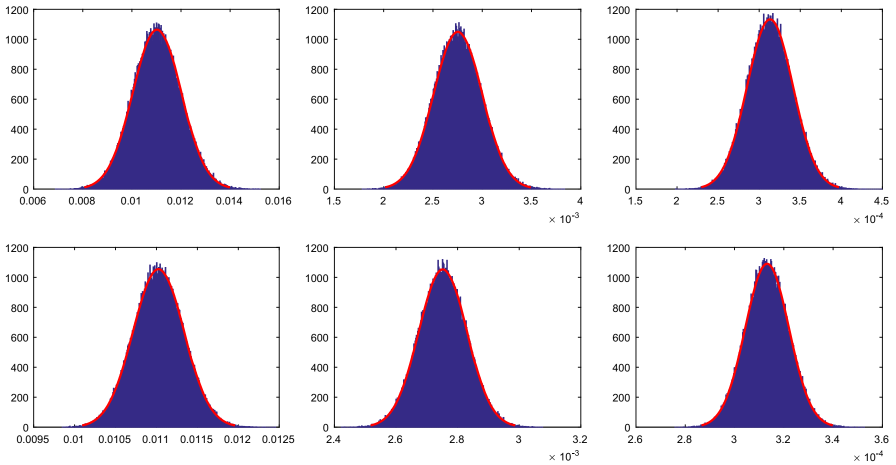

Fig. 4 Distributions of 100,000 realizations of the realized score $\widehat{S}_n^{(e)}$ for a uniform i.i.d. sample with $n = 100$ (upper part) and $n = 1000$ (lower part), for $\alpha = 0.05, 0.01, 0.001$

consequence, we will adopt one-sided tests for realized scores and two-sided tests for realized identification functions. The aim of the experiment is to compare the empirical power of the different tests in a situation in which the true data generating process for Y_i is known and the forecasting model is misspecified. The setting is very simple: we assume that the true data generating process is given by

$$Y_i = N(\mu_i, 1),\ i = 1, \dots, 100,$$

with Y_i independent and with $\mu_i = \mu^A, \mu^B, \mu^C$ (see Fig. 7). The vector of conditional means μ^A, μ^B, μ^C have been randomly selected at the beginning of the experiment

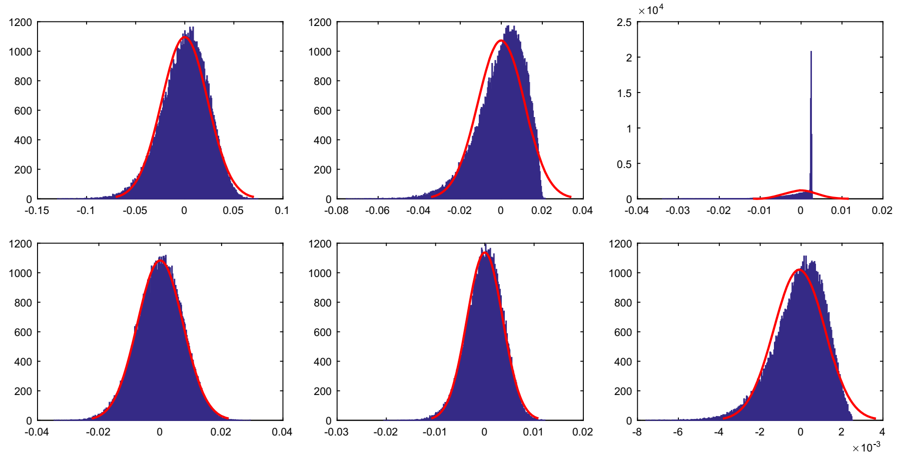

Fig. 5 Distributions of 100,000 realizations of the realized identification function $\widehat{I}_n^{(e)}$ for a normal i.i.d. sample with $n = 100$ (upper part) and $n = 1000$ (lower part), for $\alpha = 0.05, 0.01, 0.001$

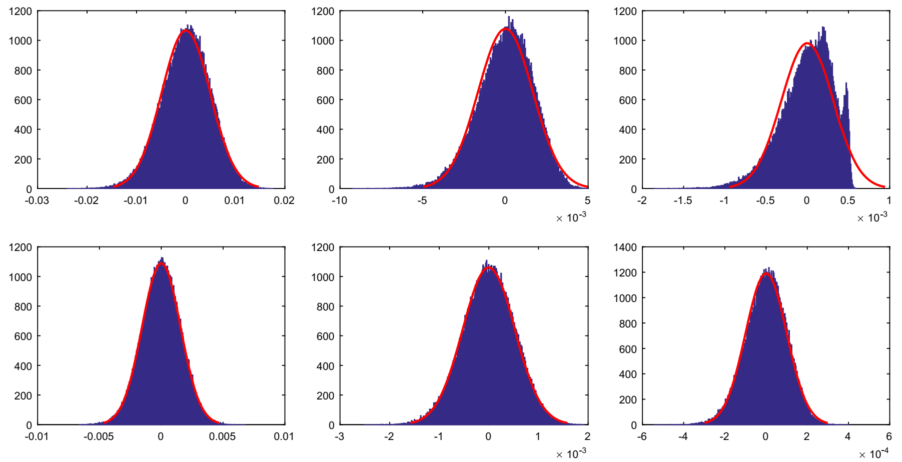

Fig. 6 Distributions of 100,000 realizations of the realized identification function $\widehat{I}_n^{(e)}$ for a uniform i.i.d. sample with $n = 100$ (upper part) and $n = 1000$ (lower part), for $\alpha = 0.05, 0.01, 0.001$

and are kept fixed throughout it. The corresponding three models will be labelled simply A, B, C. The misspecified forecasting model is

$$Y_i = N(0, 1),$$

with Y_i independent. In other words, we consider a risk manager who wrongly believes that the conditional mean is equal to 0, while in the true data generating process it has a deterministic dynamic represented by the vectors μ^A, μ^B, μ^C. We adopt such a simple simulation setting in order to show the feasibility of backtesting with realized scores and realized identification functions in the simplest possible conditions; more complex

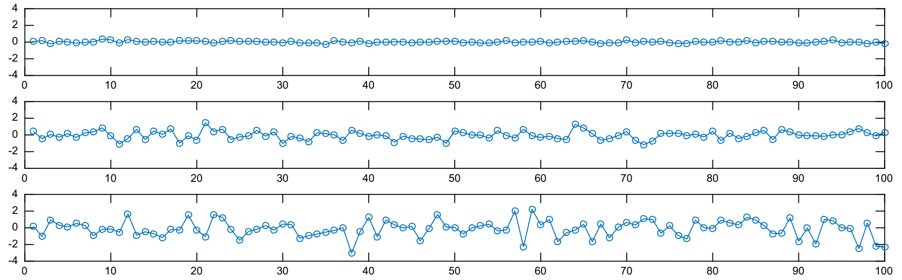

Fig. 7 Vectors of time-varying means μ^A, μ^B and μ^C

situations in which the true data generating process has an autoregressive structure or in which there is also misspecification in the variance can also be considered by a similar methodology. Here, VaR and expectile forecasts are constant and equal to the corresponding values for a standard normal: $z_{0.05} = -1.6449$ and $e_{0.05} = -1.1402$.

We start with VaR forecasts. We compare backtesting with the number of violations (that is equivalent to backtesting with the realized identification function) with backtesting with the realized score. Under the null hypothesis that the forecasting model is correct, the number of violations N_V has a binomial distribution with $p = 0.05$ and $n = 100$. Since the number of violations is an integer variable, we cannot fix a threshold such that the level of the test is exactly equal to 5%, as it would be customary. So, we decided to reject if $N_V \leq 1$ or $N_V \geq 10$, that corresponds to a level of the binomial test

$$P(N_V \leq 1) + P(N_V \geq 10) = 0.03708 + 0.02819 = 6527\%.$$

As discussed before, we use a bilateral test, rejecting either if the number of violations is too low, or if the number of violation is too big. We penalize a model if it is too permissive (too many violations), but also if it is too restrictive (too few violations). We believe that the aim of accurate risk management should not to avoid violations at all (an objective that could require excessively high margins, perhaps restricting too severely banking activity), but to keep the number of violations under control from a statistical point of view. The threshold $\overline{S}$ of the test based on the realized score $\widehat{S}_{100}^{(v)}$ is chosen by

$$P\left(\widehat{S}_{100}^{(v)} > \overline{S}\right) = 6527\%.$$

Using the asymptotic distribution derived in Sect. 3, we would get

$$\overline{S} = 0.1202.$$

However, since as we saw the distribution of $\widehat{S}_{1000.05}^{(v)}$ shows a considerable departure from normality, we prefer to use a simulated finite-sample distribution, that gives

$$\overline{S} = 0.1216.$$

Table 4 Fraction of rejections of the correct model $\mu = 0$ and of the wrong models $\mu = \mu_A$, $\mu = \mu_B$, $\mu = \mu_C$ when backtesting with the number of violations (R1) and when backtesting with scoring functions (R2)

	$\mu = 0$ (%)	$\mu = \mu_A$ (%)	$\mu = \mu_B$ (%)	$\mu = \mu_C$ (%)
$R1$	6.56	6.44	17.71	80.56
$R2$	6.66	6.98	41.76	97.60

Table 5 Fraction of rejections of the correct model $\mu = 0$ and of the wrong models $\mu = \mu_A$, $\mu = \mu_B$, $\mu = \mu_C$ when backtesting with realized identification function (R1) and when backtesting with realized score (R2)

	$\mu = 0$ (%)	$\mu = \mu_A$ (%)	$\mu = \mu_B$ (%)	$\mu = \mu_C$ (%)
$R1$	5.00	4.98	23.11	98.71
$R2$	5.01	5.59	37.00	99.87

In order to compare the empirical power of the test based on the number of violations with the test based on the realized score, we simulate $N = 100{,}000$ times models A, B and C, and as a check also the model with $\mu_i = 0\ \forall i$. We report in Table 4 the number of rejections of the forecasting model based on the number of violations (R1) and the number of rejections based on the empirical score (R2). The results show that backtesting with realized score has more power than backtesting with the number of violations in detecting the misspecification in the conditional mean. This seems quite intuitive, since the realized score contains also information on the magnitude of the violation, not only about the fact that a violation occurred.

We now consider the case of expectiles. Our aim is now to compare backtesting with the empirical identification function $\widehat{I}_n^{(e)}$ with backtesting with the empirical score $\widehat{S}_n^{(e)}$. Since now both statistics have a continuous distribution, we fix a 5% level of the test. For the identification function we consider a bilateral test with thresholds I_L and I_U determined by

$$P\left(\widehat{I}_{100}^{(e)} < I_L\right) = 2,5\% \quad \text{and} \quad P\left(\widehat{I}_{100}^{(e)} > I_U\right) = 2,5\%,$$

and for the realized score we find $\overline{S}$ such that

$$P\left(\widehat{S}_{100}^{(e)} > \overline{S}\right) = 5\%.$$

As before, since the asymptotic distributions of $\widehat{I}_n^{(e)}$ and $\widehat{S}_n^{(e)}$ show a considerable departure from normality, we determine the values of I_L, I_U and $\overline{S}$ by simulation, obtaining

$$I_L = -0.0505, \quad I_U = 0.0411, \quad \overline{S} = 0.2184.$$

The empirical powers of the two tests are reported in Table 5. As in the case of VaR, backtesting with scoring functions seems to have more power against misspecification in the conditional mean then backtesting with identification functions. We stress that this is just a toy example that may be enlarged in several directions, although it seems to indicate a general principle that backtesting with scoring function is more sensitive than backtesting with identification functions, at least with regard to misspecified conditional means.

5 A real data example

In this section we consider backtesting of 5%-VaR forecasts and 1%-expectile forecasts on real data, obtained by using as a forecasting model an AR(1)–Garch(1,1) with normal or Student's t innovations. The dataset is composed by the 1501 closing values of the SP500 Index from 03/01/2007 to 14/12/2012, from which we compute the corresponding series of 1500 daily logreturns. We estimate AR(1)–Garch(1,1) models with normal innovations and AR(1)–Garch(1,1) models with t innovations on 1000 rolling windows of 500 logreturns each; e.g. the first window uses the logreturns of days 1–500 to provide a distributional forecast for day 501, the last window uses the logreturns of days 1000–1499 to provide a distributional forecast for day 1500. At the end we get 1000 distributional forecasts, related to days 501–1500. Estimations have been performed by the Econometrics Toolbox in Matlab; we noticed some numerical instabilities in the estimation of the degrees of freedom of the innovations in the t case. The problem has been overcome by occasionally switching the default settings of Matlab's optimizer to the interior-point algorithm. The forecasted mean and standard deviations of the two models are reported in Figs. 8 and 9, while forecasted VaR and expectiles are reported in Figs. 10 and 11. The forecasted mean and standard deviations in Figs. 8 and 9 show a roughly similar behaviour, while VaR and expectiles computed

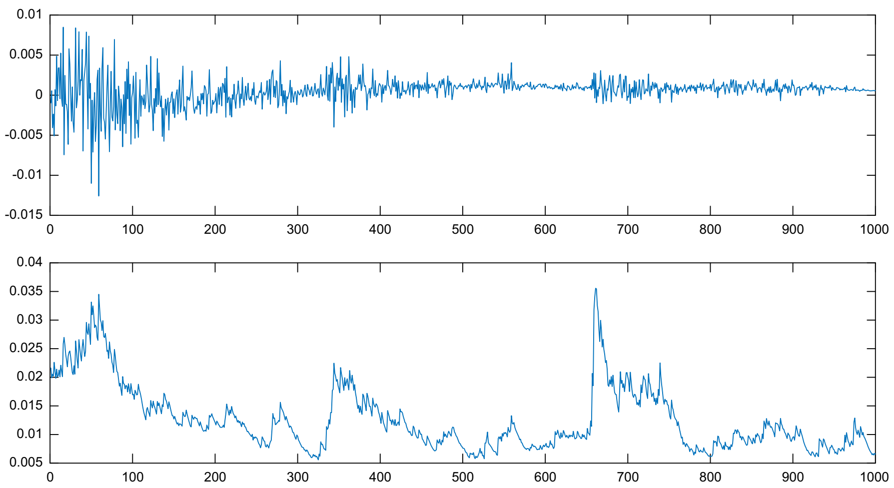

Fig. 8 Forecasted mean (upper part) and standard deviations (lower part) of the AR(1)–Garch(1,1) model with normal innovations

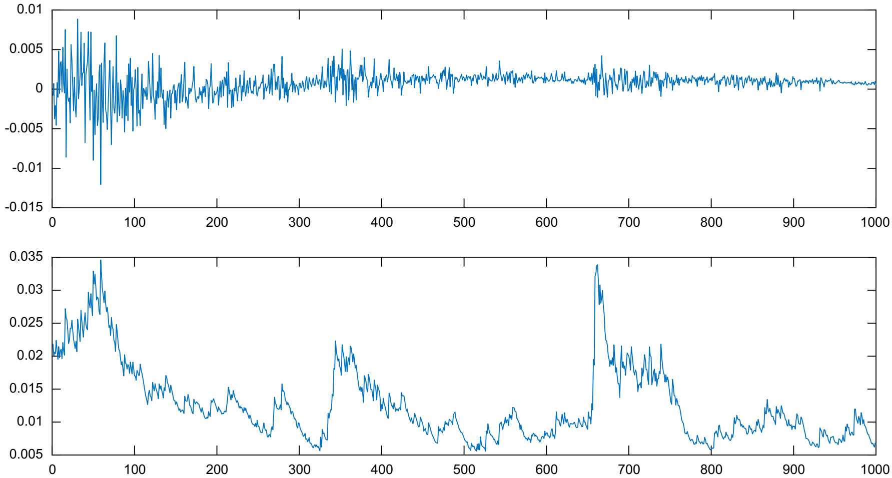

Fig. 9 Forecasted mean (upper part) and standard deviations (lower part) of the AR(1)–Garch(1,1) model with t innovations

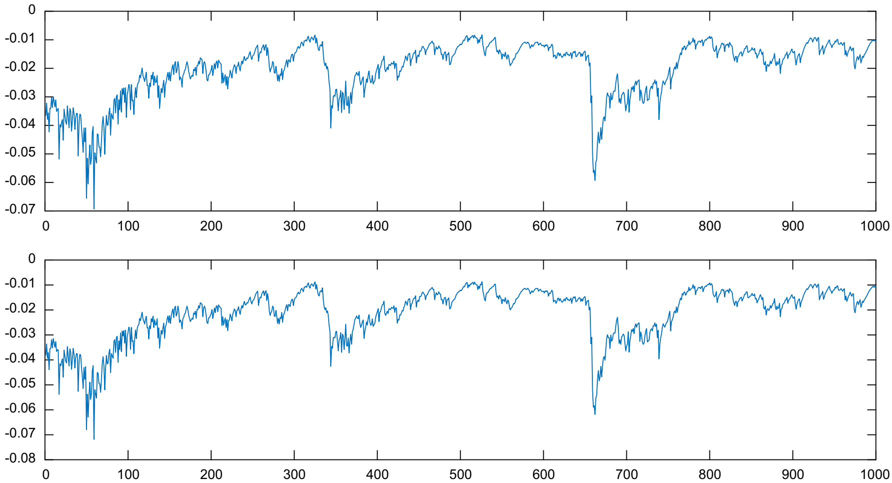

Fig. 10 Forecasted 5% VaR (upper part) and 1% expectiles (lower part) of the AR(1)–Garch(1,1) model with normal innovations

by t model in Fig. 11 are typically slightly lower than the corresponding quantities computed with the normal model in Fig. 10, as a consequence of the heavier tails.

Our aim is now to backtest these forecasts by means of realized scores and identification functions. The problem is that in the considered Garch models the logreturns are not independent, so clearly the results derived in Sect. 3 on asymptotic distribution of realized scores and identification functions are no longer applicable. In order to circumvent this issue, we suggest two possible approaches. In the first one, we determine the empirical distribution of the realized score under the null hypothesis that the

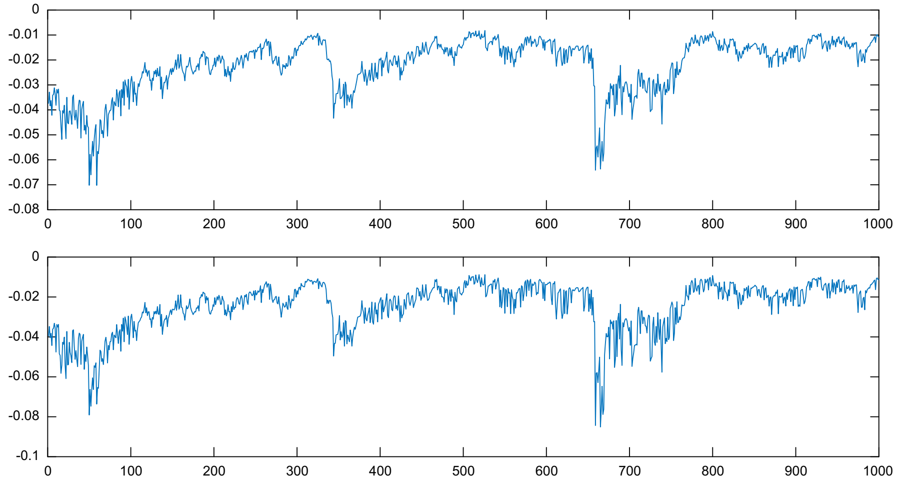

Fig. 11 Forecasted 5% VaR (upper part) and 1% expectiles (lower part) of the AR(1)–Garch(1,1) model with t innovations

forecasting model is correct by means of simulations, as in Bellini and Di Bernardino (2017).

In the second one we first perform a probability integral transform (PIT) on the realized logreturns $U_i = F_{Y_i}(Y_i)$. Indeed, under the null that the forecasting model is correct, the resulting sequence U_i is uniform i.i.d., so we can apply the asymptotic results derived in Sect. 3. A similar idea has been suggested by Kerkhof and Melenberg (2004) for backtesting Expected Shortfall.

5.1 Backtesting by simulating realized scores

In order to derive the distribution of the realized scores $\widehat{S}^{(v)}_{1000}$ and $\widehat{S}^{(e)}_{1000}$ under an AR(1)–Garch(1,1) model with normal or t innovations, we simulate $N = 100{,}000$ trajectories of length $n = 1000$. For each day, we simulate a logreturn from the forecasting distribution obtained by the estimated Garch model, and we compute the corresponding scoring function; realized scores are then time averages of these quantities. Notice that in the normal case the forecasting distribution is specified by two parameters, while in the t case also the estimated degrees of freedom is used, typically varying in the range 4–20.

In Table 6 we report the results of backtesting. We see that in the normal case the realized value of the scoring functions $\widehat{S}^{(v)}_{1000}$ and $\widehat{S}^{(e)}_{1000}$ are quite high with respect to their mean under H_0, leading to a rejection at the 5%-level with p values respectively equal to 0.00072 and 0.02164. On the contrary the model with t innovations is not rejected at the same level, since the p values are respectively equal to 0.52471 and 0.6136. These findings are consistent with the usual “stylized fact” of conditional heavy-tailedness.

Table 6 Backtesting of the normal and t model with realized scores

	Normal case		t case	
	$\widehat{S}^{(v)}_{1000}$	$\widehat{S}^{(e)}_{1000}$	$\widehat{S}^{(v)}_{1000}$	$\widehat{S}^{(e)}_{1000}$
Realized value	1.5044×10^{-3}	1.3790×10^{-5}	1.4883×10^{-3}	1.2826×10^{-5}
Mean under H_0	1.3081×10^{-3}	1.0047×10^{-5}	1.5157×10^{-3}	1.5802×10^{-5}
SD under H_0	4.9517×10^{-5}	1.0969×10^{-6}	8.0590×10^{-5}	1.0063×10^{-5}
p value	0.00072	0.02164	0.52471	0.6136

In the first line we report the realized value of the test statistic, in the second and third line the mean and the standard deviation under the null, and in the fourth line the p value, computed by means of risimulations as described in Sect. 5.1

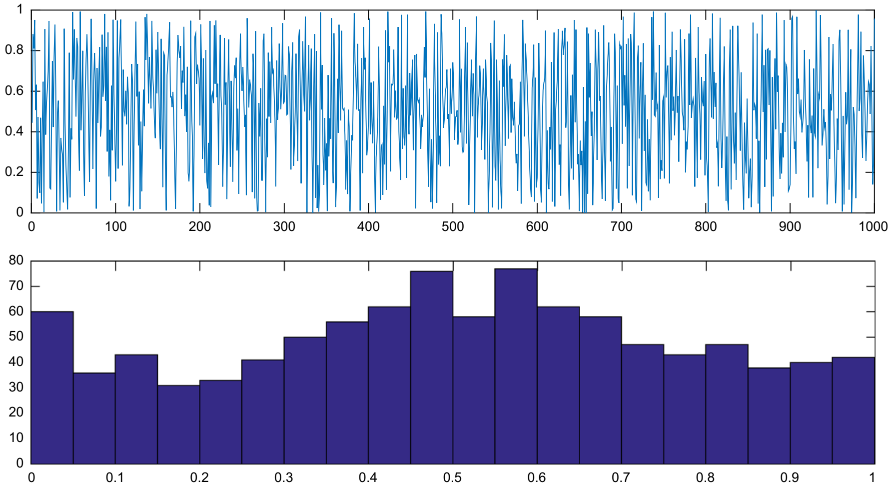

Fig. 12 Time series (upper part) and distribution (lower part) of the probability integral transform in the AR(1)–Garch(1,1) model with normal innovations

5.2 Backtesting by probability integral transform

In order to implement the second backtesting approach, we first compute the PIT for the normal and Student's t estimated models. The time series of the PIT and their distributions are reported in Fig. 12 for the normal case and in Fig. 13 for the t case. A first graphical inspection shows a reasonable agreement with the uniform i.i.d. hypothesis. We now compute the realized identification functions and realized scores as if the model would be uniform i.i.d., and we compute the thresholds for the corresponding tests by means of the asymptotic results in Sect. 3 for the uniform i.i.d. case.

In Table 7 we report the results of backtesting. Notice that in this approach the distribution of the various test statistics under the null is the same in the normal and in the t model, since in both cases the PIT is uniform i.i.d.

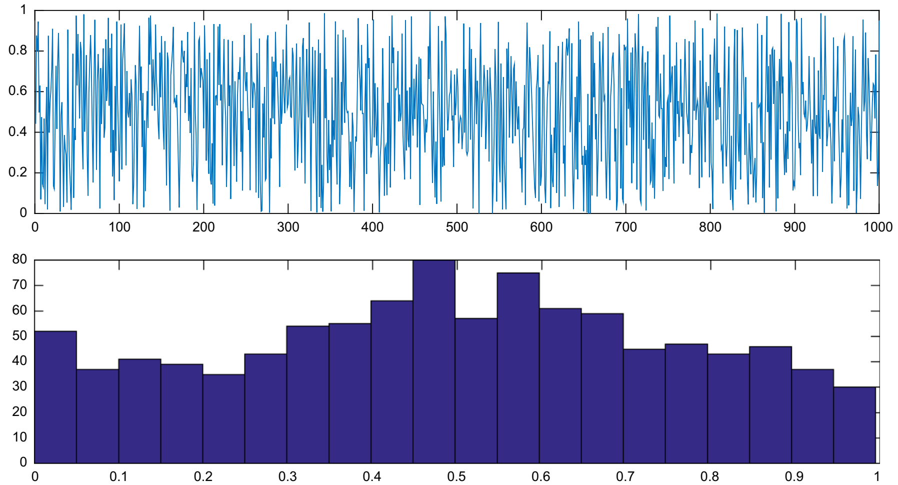

Fig. 13 Time series (upper part) and distribution (lower part) of the probability integral transform in the AR(1)–Garch(1,1) model with t innovations

Table 7 Backtesting of the normal and t model with realized scores and identification functions, computed after applying the PIT

	N_V	$\widehat{S}^{(v)}_{1000}$	$\widehat{I}^{(e)}_{1000}$	$\widehat{S}^{(e)}_{1000}$
Normal case				
Realized value	60	2.4497×10^{-2}	-8.7402×10^{-4}	2.7603×10^{-3}
Mean under H_0	50	2.3750×10^{-2}	0	2.7523×10^{-3}
SD under H_0	6.8920	4.3361×10^{-4}	5.2331×10^{-4}	7.7847×10^{-5}
p value	0.0734	0.0425	0.04744	0.45907
t case				
Realized value	52	2.3801×10^{-2}	-2.1496×10^{-4}	2.6088×10^{-3}
Mean under H_0	50	2.3750×10^{-2}	0	2.7523×10^{-3}
SD under H_0	6.8920	4.3361×10^{-4}	5.2331×10^{-4}	7.7847×10^{-5}
p value	0.3858	0.4532	0.34062	0.96736

In the first line we report the realized value of the test statistic, in the second and third line the mean and the standard deviation under the null, and in the fourth line the p value, computed by means of the asymptotic distribution for the uniform i.i.d. case derived in Sect. 3

Starting from the VaR forecasts, we see that the number of violations N_V in the normal case exceeds its mean under H_0, but this does not lead to a rejection (p value equal to 0.0734). But if we look at the realized score, we have a rejection of the normal model with p value equal to 0.0425. This shows that backtesting with realized scores can reject in situations where standard backtesting with number of violations would not. In the t case VaR forecasts are not rejected either by number of violations or by the realized score.

The situation is similar for expectile forecasts. In the normal case they are rejected by the test based on realized identification function, while in the t case they are not rejected by both tests.

5.3 Discussion

For the sake of clarity, we stress that this is not a comparative backtesting as in Nolde and Ziegel (2016), based on a Diebold–Mariano test on difference of realized scores, but we perform a separate backtest of the two models.

In the case of expectiles the opposite situation arise: the normal model would be rejected by backtesting with the realized identification function but would pass the test based on the realized score. Indeed, the t-model performs much better in terms of realized score, but the performance of the normal model is not as bad as to produce a rejection.

6 Conclusions and directions for further research

The provided simulated and real data examples show that in principle it may be sensible to implement a backtesting procedure based on the realized value of the scoring function, with the idea of rejecting the model if the realized score is too big. Apparently, this approach captures the idea of backtesting by keeping into account also the magnitude of violations and not only their number; it may have more power than the traditional one, at least in simple examples.

Our use of the scoring function is different from Fissler et al. (2016) and Nolde and Ziegel (2016), since we are not testing hypotheses relative to the comparative merit of two models, but we are backtesting a single model, as in the traditional Basel I framework. From a statistical point of view our approach requires the determination of the distribution of the realized scoring function under the null hypothesis that the forecasting model is correct; we compute the asymptotic distribution for i.i.d. normal and uniform samples, and we use simulations for more complicated econometric models.

As directions for further research, we believe it would be important to find a more accurate approximation for finite-sample distributions of realized scores. Finally, the same ideas may be applied to the traditional backtesting of the couple (VaR, ES) by means of the consistent scoring functions studied in Acerbi and Szekely (2014) and Fissler and Ziegel (2016).

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Appendix

We report in this Appendix the explicit computation of the expected values and variances of piecewise linear and quadratic scores and identification function in the normal and uniform cases. That is, we compute the quantities $E[S^{(v)}(x, Y)]$, $E\left[S^{(e)}(x, Y)\right]$,

$E[I^{(v)}(x,Y)]$, $E[I^{(e)}(x,Y)]$ and the corresponding variances when $Y \sim N(0,1)$ and $Y \sim U(0,1)$. We denote with $\phi(x)$, $\Phi(x)$ and $\overline{\Phi}(x)$ the density, the cumulative and the retro-cumulative function of a standard normal r.v.

The following identities will be repeatedly used:

$$\int_{-\infty}^{x} y\phi(y)dy = -\phi(x), \int_{x}^{+\infty} y\phi(y)dy = \phi(x)$$
$$\int_{-\infty}^{x} y^2\phi(y)dy = \Phi(x) - x\phi(x), \int_{x}^{+\infty} y^2\phi(y)dy = \overline{\Phi}(x) + x\phi(x)$$
$$\int_{-\infty}^{x} y^3\phi(y)dy = -x^2\phi(x) - 2\phi(x), \int_{x}^{+\infty} y^3\phi(y)dy = x^2\phi(x) + 2\phi(x)$$
$$\int_{-\infty}^{x} y^4\phi(y)dy = -x^3\phi(x) + 3\left[\Phi(x) - x\phi(x)\right]$$
$$\int_{x}^{+\infty} y^4\phi(y)dy = x^3\phi(x) + 3\left[\overline{\Phi}(x) + x\phi(x)\right]$$

Piecewise linear score, normal case

$$\begin{aligned}
E[S^{(v)}(x,Y)] &= \alpha\int_{x}^{+\infty}(y-x)\phi(y)dy + (1-\alpha)\int_{-\infty}^{x}(x-y)\phi(y)dy \\
&= \alpha[\phi(x) - x\overline{\Phi}(x)] + (1-\alpha)[x\Phi(x) + \phi(x)] \\
&= \phi(x) + x[\Phi(x) - \alpha].
\end{aligned}$$

When $x = z_\alpha$, we get

$$\begin{aligned}
E[S^{(v)}(z_\alpha, Y)] &= \phi(z_\alpha). \\
E\left[S^{(v)}(x,Y)^2\right] &= \int_{-\infty}^{+\infty}\left[\alpha(y-x)_+ + (1-\alpha)(y-x)_-\right]^2\phi(y)dy \\
&= \alpha^2\int_{x}^{+\infty}(y-x)^2\phi(y)dy + (1-\alpha)^2\int_{-\infty}^{x}(y-x)^2\phi(y)dy \\
&= \alpha^2\left\{\overline{\Phi}(x) - x\phi(x) + x^2\overline{\Phi}(x)\right\} \\
&\quad + (1-\alpha)^2\left\{\Phi(x) + x\phi(x) + x^2\Phi(x)\right\} \\
&= \left[1 + x^2\right]\left\{\alpha^2 + (1-2\alpha)\Phi(x)\right\} + (1-2\alpha)x\phi(x).
\end{aligned}$$

When $x = z_\alpha$, we get

$$\begin{aligned}
E\left[S^{(v)}(x,Y)^2\right] &= \left[1 + z_\alpha^2\right]\left\{\alpha - \alpha^2\right\} + (1-2\alpha)z_\alpha\phi(z_\alpha) \\
Var[(S^{(v)}(z_\alpha, Y)] &= E\left[S^{(v)}(x,Y)^2\right] - \phi^2(z_\alpha) \\
&= \alpha(1-\alpha)(1+z_\alpha^2) + (1-2\alpha)z_\alpha\phi(z_\alpha) - \phi^2(z_\alpha).
\end{aligned}$$

Piecewise quadratic score, normal case

$$
\begin{aligned}
E\left[S^{(e)}(x,Y)\right] &= \alpha\int_x^{+\infty}(y-x)^2\phi(y)dy + (1-\alpha)\int_{-\infty}^{x}(x-y)^2\phi(y)dy \\
&= \alpha[\overline{\Phi}(x) + x\phi(x) + x^2\overline{\Phi}(x) - 2x\phi(x)] \\
&\quad + (1-\alpha)[\Phi(x) - x\phi(x) + x^2\Phi(x) + 2x\phi(x)] \\
&= (1-2\alpha)x\phi(x) + (1+x^2)[\alpha\overline{\Phi}(x) + (1-\alpha)\Phi(x)]. \\
E\left[S^{(e)}(x,Y)^2\right] &= \int_{-\infty}^{+\infty}\left[\alpha(y-x)_+^2 + (1-\alpha)(y-x)_-^2\right]^2\phi(y)dy \\
&= \int_x^{+\infty}\alpha^2(y-x)^4\phi(y)dy + \int_{-\infty}^{x}(1-\alpha)^2(y-x)^4\phi(y)dy.
\end{aligned}
$$

Now we have for the first term:

$$
\begin{aligned}
\int_x^{+\infty}(y-x)^4\phi(y)dy &= x^3\phi(x) + 3\left[\overline{\Phi}(x) + x\phi(x)\right] - 4x\left[x^2\phi(x) + 2\phi(x)\right] \\
&\quad + 6x^2\left[\overline{\Phi}(x) + x\phi(x)\right] - 4x^3\phi(x) + x^4\overline{\Phi}(x) \\
&= \overline{\Phi}(x)\left[3 + 6x^2 + x^4\right] - 5x\phi(x) - x^3\phi(x).
\end{aligned}
$$

Similarly for the second term:

$$
\begin{aligned}
\int_{-\infty}^{x}(y-x)^4\phi(y)dy &= -x^3\phi(x) + 3\left[\Phi(x) - x\phi(x)\right] - 4x\left[-x^2\phi(x) - 2\phi(x)\right] \\
&\quad + 6x^2\left[\Phi(x) - x\phi(x)\right] - 4x^3\left[-\phi(x)\right] + x^4\Phi(x) \\
&= \Phi(x)\left[3 + 6x^2 + x^4\right] + x^3\phi(x) + 5x\phi(x).
\end{aligned}
$$

Summing up, we get

$$
\begin{aligned}
E\left[S^{(e)}(x,Y)^2\right] &= \left(3 + 6x^2 + x^4\right)\left(\alpha^2 - 2\alpha\Phi(x) + \Phi(x)\right) \\
&\quad + (1-2\alpha)\left(5x\phi(x) + x^3\phi(x)\right)
\end{aligned}
$$

When $x = e_\alpha$, we get

$$
\begin{aligned}
Var\left[S^{(e)}(e_\alpha,Y)\right] &= \left(3 + 6e_\alpha^2 + e_\alpha^4\right)\left(\alpha^2 - 2\alpha\Phi(e_\alpha) + \Phi(e_\alpha)\right) \\
&\quad + (1-2\alpha)\left(5e_\alpha\phi(e_\alpha) + e_\alpha^3\phi(e_\alpha)\right) + \\
&\quad - \left((1-2\alpha)e_\alpha\phi(e_\alpha) + \left(1 + e_\alpha^2\right)\left(\alpha\overline{\Phi}(e_\alpha) + (1-\alpha)\Phi(e_\alpha)\right)\right)^2.
\end{aligned}
$$

Identification function, normal case. Recall that

$$I^{(e)}(Y,e) = \alpha(Y-e)_+ - (1-\alpha)(Y-e)_-.$$

Since $E[I^{(e)}(Y,e_\alpha)] = 0$, we get

$$\begin{aligned}
Var[I^{(e)}(Y,e_\alpha)] &= \alpha^2 \int_\varepsilon^{+\infty} (y-e_\alpha)^2 \phi(y)dy + (1-\alpha)^2 \int_{-\infty}^{e_\alpha} (e_\alpha - y)^2 \phi(y)dy \\
&= \alpha^2 \left[\overline{\Phi}(e_\alpha) + e_\alpha \phi(e_\alpha) - 2e_\alpha \phi(e_\alpha) + e_\alpha^2 (1-\Phi(e_\alpha)) \right] \\
&\quad + (1-\alpha)^2 \left[\Phi(e_\alpha) - e_\alpha \phi(e_\alpha) + 2e_\alpha \phi(e_\alpha) + e_\alpha^2 \Phi(e_\alpha) \right] \\
&= (1+e_\alpha^2) \left[\alpha^2 + \Phi(e_\alpha)(1-2\alpha) \right] + e_\alpha \phi(e_\alpha) \left[1-2\alpha\right].
\end{aligned}$$

Let now $Y \sim U(0,1)$, $\phi(y) = 1_{[0,1]}$, $\Phi(y) = y 1_{[0,1]}$ and $z_\alpha = \alpha$.

Piecewise linear score, uniform case.

$$\begin{aligned}
E[S^{(v)}(z_\alpha, Y)] &= \alpha \int_\alpha^1 (y-\alpha)\phi(y)dy + (1-\alpha) \int_0^1 (\alpha - y)\phi(y)dy \\
&= \frac{\alpha(1-\alpha)}{2}. \\
E[S^{(v)}(z_\alpha, Y)^2] &= \alpha^2 \int_\alpha^1 (y-\alpha)^2 dy + (1-\alpha)^2 \int_0^\alpha (\alpha - y)^2 dy \\
Var[S^{(v)}(z_\alpha, Y)] &= E[S^{(v)}(z_\alpha, Y]^2) - [E[S^{(v)}(z_\alpha, Y)]]^2 \\
&= \frac{1}{12}\alpha^2 (\alpha - 1)^2
\end{aligned}$$

Piecewise quadratic score, uniform case. Recall that if $Y \sim U(0,1)$

$$e_\alpha = \frac{\alpha - \sqrt{\alpha - \alpha^2}}{2\alpha - 1}. \tag{9}$$

$$\begin{aligned}
E\left[S^{(e)}(e_\alpha, Y)\right] &= \int_0^1 \left[\alpha(y-e_\alpha)_+^2 + (1-\alpha)(y-e_\alpha)_-^2 \right] dy \\
&= \alpha \int_{e_\alpha}^1 (y-e_\alpha)^2 dy + (1-\alpha) \int_0^{e_\alpha} (e_\alpha - y)^2 dy \\
&= \frac{1}{3} e_\alpha^3 (1-\alpha) + \alpha \left(e_\alpha^2 - e_\alpha - \frac{1}{3} e_\alpha^3 + \frac{1}{3} \right)
\end{aligned}$$

Substituting (9) we get

$$E\left[S^{(e)}(e_\alpha, Y)\right] = \frac{1}{3}\alpha(1-\alpha)\frac{1-2\sqrt{\alpha(1-\alpha)}}{(2\alpha-1)^2}.$$

$$\begin{aligned}
E[S^{(e)}(e_\alpha, Y)^2] &= \int_0^1 \left[\alpha(y-e_\alpha)_+^2 + (1-\alpha)(y-e_\alpha)_-^2\right]^2 dy \\
&= \alpha^2 \int_{e_\alpha}^1 (y-e_\alpha)^4 dy + (1-\alpha)^2 \int_0^{e_\alpha} (e_\alpha - y)^4 dy \\
&= \frac{1}{5}e_\alpha^5\,(1-\alpha)^2 + \alpha^2\left(2e_\alpha^2 - e_\alpha - 2e_\alpha^3 + e_\alpha^4 - \frac{1}{5}e_\alpha^5 + \frac{1}{5}\right)
\end{aligned}$$

$$\begin{aligned}
E[S^{(e)}(e_\alpha, Y)^2] &= -\frac{11\alpha^4 - 2\alpha^3 - \alpha^2 - 12\alpha^5 + 4\alpha^6 - \left(\alpha-\alpha^2\right)^{\frac{5}{2}}}{5\,(2\alpha-1)^4} + \\
&\quad -\frac{5\alpha^2\sqrt{\alpha-\alpha^2} - 10\alpha^3\sqrt{\alpha-\alpha^2} + 5\alpha^4\sqrt{\alpha-\alpha^2}}{5\,(2\alpha-1)^4} \\
&= -\frac{1}{5}\alpha^2\left(-4\alpha + 4\alpha^2 - 1\right)\frac{(\alpha-1)^2}{(2\alpha-1)^4} - \frac{4}{5}\frac{\left(\alpha-\alpha^2\right)^{\frac{5}{2}}}{(2\alpha-1)^4} \\
&= \frac{\alpha^2\,(\alpha-1)^2}{5\,(2\alpha-1)^4}\left(4\alpha - 4\alpha^2 + 1 - 4\sqrt{\alpha-\alpha^2}\right) \\
&= \frac{\alpha^2\,(1-\alpha)^2}{5\,(2\alpha-1)^4}\left(1-2\sqrt{\alpha(1-\alpha)}\right)^2.
\end{aligned}$$

$$\begin{aligned}
Var[S^e(e_\alpha, Y)] &= \frac{\alpha^2\,(1-\alpha)^2}{5\,(2\alpha-1)^4}\left(1-2\sqrt{\alpha(1-\alpha)}\right)^2 + \\
&\quad -\left(\frac{1}{3}\alpha(1-\alpha)\frac{1-2\sqrt{\alpha(1-\alpha)}}{(2\alpha-1)^2}\right)^2 \\
&= \frac{4}{45}\frac{\alpha^2\,(1-\alpha)^2}{(2\alpha-1)^4}\left(1-2\sqrt{\alpha(1-\alpha)}\right)^2
\end{aligned}$$

Note the relationship

$$SD[S^e(e_\alpha, Y)] = \frac{2}{3\sqrt{5}}\frac{\alpha\,(1-\alpha)}{(2\alpha-1)^2}\left(1-2\sqrt{\alpha(1-\alpha)}\right) = \frac{2}{\sqrt{5}}E[S^e(e_\alpha, Y)].$$

Identification function, uniform case.

$$\begin{aligned}
Var[I^e(Y, e_\alpha)] &= \alpha^2 \int_{e_\alpha}^1 (y-e_\alpha)^2 dy + (1-\alpha)^2 \int_0^{e_\alpha} (y-e_\alpha)^2 dy \\
&= \frac{\alpha^2}{3}\left(3e_\alpha^2 - 3e_\alpha - e_\alpha^3 + 1\right) + (1-\alpha)^2\frac{1}{3}e_\alpha^3 \\
&= \frac{\alpha^2}{3}\,(1-e_\alpha)^3 + \frac{1}{3}(1-\alpha)^2 e_\alpha^3.
\end{aligned}$$

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